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Order Finding and Factoring

Order Finding on a Quantum Computer

Fast Fourier Transform: Brief Recap from Course # 3

Quantum Fourier Transform

Video: Quantum Fourier Transform: Part 1
18 min

Video: Quantum Fourier Transform: Part 2
28 min

Lab: Interactive Detailed Notes on Quantum Fourier Transform
1h

Graded Assignment: Quantum Fourier Transform
1h

Problem Set Week # 3

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Quantum Fourier Transform

Review Learning Objectives

1. Which of the matrices below is the correct one for 2 qubit QFT. **2 / 2 points**

Note that $\omega_4 = \exp(\frac{2\pi i}{4}) = i$

- $\frac{1}{2}\begin{pmatrix}1&1&1&1\\1&0&-1&0\\1&-1&1&-1\\1&0&-1&0\end{pmatrix}$
- $\frac{1}{2}\begin{pmatrix}1&1&1&1\\1&i&-1&-i\\1&-1&1&-1\\1&-i&-1&i\end{pmatrix}$ ✓
- $\frac{1}{\sqrt{2}}\begin{pmatrix}1&1&1&1\\1&i&-1&-i\\1&-1&1&-1\\1&-i&-1&i\end{pmatrix}$
- $\frac{1}{\sqrt{2}}\begin{pmatrix}1&1\\1&-1\end{pmatrix}$

Correct

Assignment details

Due May 19, 11:59 PM IST
Attempts Unlimited
Submitted May 7, 12:06 AM IST

Try again

Your grade
To pass you need at least 80%. We keep your highest score.

100%

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2. Suppose U is the unitary matrix for a 2-qubit QFT. Select all the true facts. **4 / 4 points**

- [x] $U|11\rangle = \frac{1}{2}(|00\rangle - i|01\rangle - |10\rangle + i|11\rangle)$

Correct

- [ ] $U|10\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)$
- [x] $U|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

Correct

- [x] $U\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$

Correct

Note that $U \times \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\\0\end{pmatrix}$

- [x] $U\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) = |00\rangle$

Correct

Note that $U \times \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$

- [ ] $U|00\rangle = U|11\rangle$

3. Suppose we would like to take the 2-bit QFT of the uniform superposition **3 / 3 points**

$|\psi\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |01\rangle + |11\rangle)$

by using the quantum divide-and-conquer method. Select all the correct answers from the list below.

- [x] The algorithm recursively applies 1 bit QFT in parallel to the states $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and $|\psi_2\rangle = |0\rangle$ since $|\psi\rangle = \frac{\sqrt{2}}{\sqrt{3}}|\psi_1\rangle \otimes |0\rangle + \frac{1}{\sqrt{3}}|\psi_2\rangle \otimes |1\rangle$

Correct

- [x] The 1 bit QFT applied to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ yields $|0\rangle$

Correct
Correct

- [x] The 1 bit QFT applied to $|0\rangle$ yields $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Correct
Correct

- [ ] The result of the QFT of $|\psi\rangle$ is therefore the sum of the QFTs of $|\psi_1\rangle$ and ψ_2 above.

- [x] If $o_0|0\rangle + o_1|1\rangle$ and $e_0|0\rangle + e_1|1\rangle$ are the one bit QFT results for the even and odd parts of the super position (assume $|o_0|^2 + |o_1|^2 + |e_0|^2 + |e_1|^2 = 1$),

the result of the QFT should be

$(o_0 + e_0)|00\rangle + (o_1 + ie_1)|01\rangle + (o_0 - e_0)|10\rangle + (o_1 - ie_1)|11\rangle$

Correct
Correct: this captures the combine step for this case.